Reaction Kinetics Visualisation with Basic Thermodynamics: Project Report

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1. Problem Statement and Theory:

A first-order reversible reaction, the gas-phase decomposition of dinitrogen peroxide

$N_2O_5(g) \rightarrow 2\ NO_2(g) + ½\ O_2(g)$

takes place in an constant-volume batch reactor of V = 2.0 L, initially containing n_0 = 0.025 mol of N_2O_5, so the initial concentration is $C_0 = n_0/V$ = 0.0125 mol/L. The reaction is first order in N_2O_5:

$dC/dt = -kC$ with $C(0) = C_0$

Where C is the concentration of A(mol/L), t is time (s), and k is the rate constant (s^{-1}). The analytical solution is $C(t) = C_0\, e^{(-kt)}$, with half-life $t_{1/2} = \ln 2/k$.

Experimental data. The first-order rate constant has been measured at four temperatures (data taken from a problem 12.3 in Keeler and Wothers (2014), *Chemical Structure and Reactivity* (2nd ed.), Oxford University Press (2nd ed))

T(K)	349	368	378	396
k(s^{-1})	0.0068	0.041	0.13	0.50

The temperature dependence of the rate constant follows the Arrhenius equation: $k(T) = A \cdot e^{(-Ea/RT)}$, with $A = 5.7 \times 10^{13}\ s^{-1}$ and activation energy

E_a = 106 kJ/mol, and R = 8.314 J/(mol K). For validation, a literature half-life of 4.03×10^4 s at 25 °C is used ($k = 1.72 \times 10^{-5}$ s^{-1}).
Thermodynamic data. Standard formation data at 298 K:

Species	ΔfH° (kJ/mol)	S° (J/(mol·K))
$N_2O_5(g)$	11.3	355.7
$NO_2(g)$	33.18	240.06
$O_2(g)$	0	205.14

The reaction enthalpy and entropy are ΔH° = Σ ΔfH°(products) − Σ ΔfH°(reactants) and ΔS° likewise, both per mole of N_2O_5. The Gibbs energy is **Δ**G°=**Δ**H°-T**Δ**S°, and the equilibrium constant is K=exp(-**Δ**G°/RT) = (p_NO_2)²(p_O_2)^½ / p_N_2O_5 with partial pressures in bar
To find:

1. **Effect of Rate constant.** Plot C(t) over 0 to 300 s for k= 0.0068, 0.041, 0.13 and 0.50 s^{-1}, on linear and semi-logarithmic axes. Mark each half-life and discuss how reaction speed changes with k.
2. **Numerical verification.** Solve dC/dt = −kC for k = 0.0068 s^{-1} with MATLAB's `ode45` (RelTol 10^{-8}, AbsTol 10^{-10}), compare it with the analytical solution, and plot the absolute error.
3. .**Arrhenius analysis.** Fit ln k against 1/T to the measured data by linear regression to obtain E_a and A, and report R^2. Extrapolate the fit to 25 °C and compare the predicted rate constant and half-life with the literature values.
4. **Thermodynamic limit.** For T from 290 to 400 K, calculate ΔG°, K, ΔH°, ΔS° and find the equilibrium conversion of pure N_2O_5 at total pressures of 1 bar and 0.1 bar. Calculate the heat absorbed by

the reacting batch, $Q(t) = n_0 \cdot \Delta H^\circ \cdot (1 - e^{(-kt)})$, at each measured temperature

2. Objective:

The objective of this project is to model and visualise how the rate constant and temperature affect the speed of a real first-order gas-phase reaction, to find its activation energy from experimental data, and compare results with independent literature data and use thermodynamics to show how far the reaction goes and how much heat it absorbs.

3. Assumptions:

- The reaction is first order in N_2O_5 at all temperatures studied
- Ideal gas behaviour and isothermal operation at each temperature
- The overall reaction is treated as a single step
- E_a and A are constant over the temperature range, and ΔH° and ΔS° are independent of temperature
- The rate data are textbook values; here we are considering the one stated above

4. Data :

Thermodynamic and kinetic data are given in section 1.
Reactor basis

Item	Symbol	Value	Unit
Reactor Volume	V	[2.0]	L
Initial amount of N_2O_5	n_0	[0.025]	mol
Initial Concentration	C_0	[0.0125]	mol/L

Gas constant	R	8.314	(J/mol K)

4. Method and Code Overview :

The code of 4 parts and beginning just states the numerical values given by the data:

- Part 1: This part plots the decay curves for the four measured rate constant in linear and semi log axes with half lives marked using basic functions and loops
- Part 2: solving the ODEs with given data using numerical methods ode45 (RelTol [1e-8], AbsTol [1e-12]) and compare it with analytical solution.
- Part 3: fits ln k against 1/T with polyfit to obtain E_a, A and R^2 then extrapolates the fit to 25 °C and compares it with the literature value.
- Part 4: Calculates the $\Delta G°$, K, $\Delta H°$, $\Delta S°$ values from given data , solves for the equilibrium conversion with fzero at 1 bar and 0.1 bar and calculates the heat absorbed.

5. Result and Graphs:

Part 1: Effect of temperature on the decay

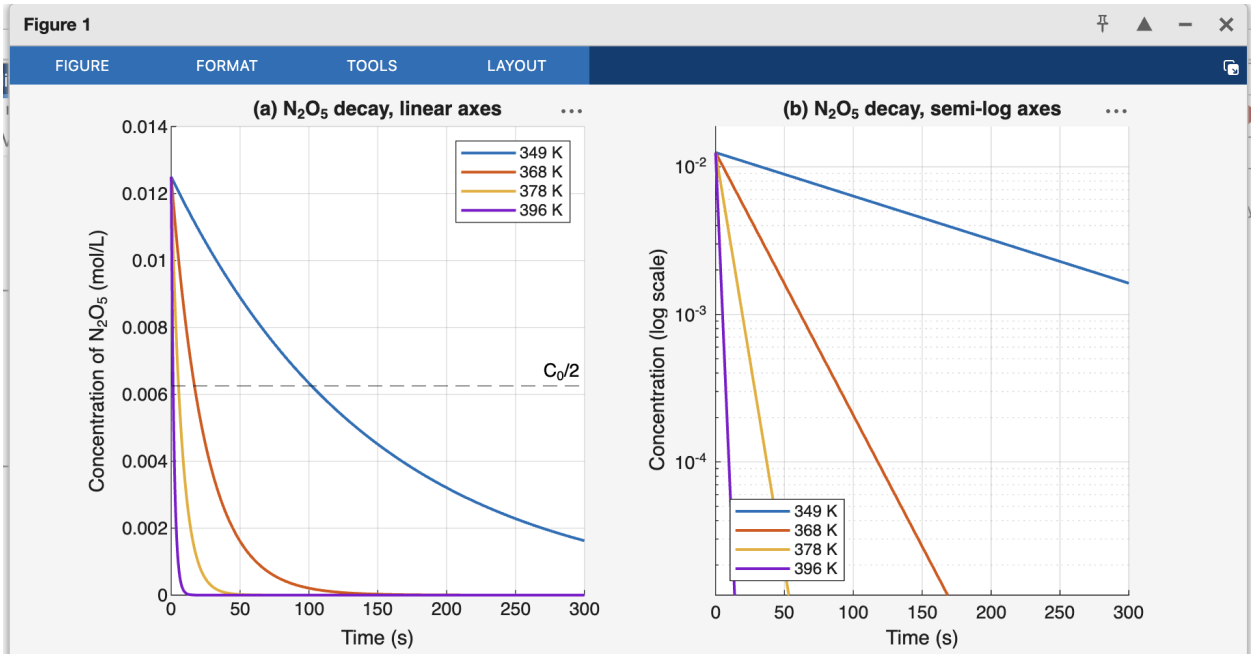


Figure 1: Plot (a) is linear and (b) is semi-log axes

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PART 1: measured rate constants and half-lives
  T (K)      k (1/s)     t_half (s)
    349       0.0068         101.93
    368       0.0410          16.91
    378       0.1300           5.33
    396       0.5000           1.39
```

This plot shows that the he concentration of N_2O_5 decays exponentially with time for all tested temperatures. Higher temperatures (e.g., $396K$) result in significantly faster decomposition rates compared to lower temperatures ($349K$), as evidenced by the sharper downward curves and shorter half-lives ($t_{1/2}$ marked at $C_0/2$) Plotting $\ln(C)$ versus time yields linear trends whose slopes equal $-\ k$. This confirms that the gas-phase decomposition of N_2O_5 strictly follows first-order kinetics, with the slope becoming steeper as temperature increases.

Part 2: Numerical solution versus analytical solution

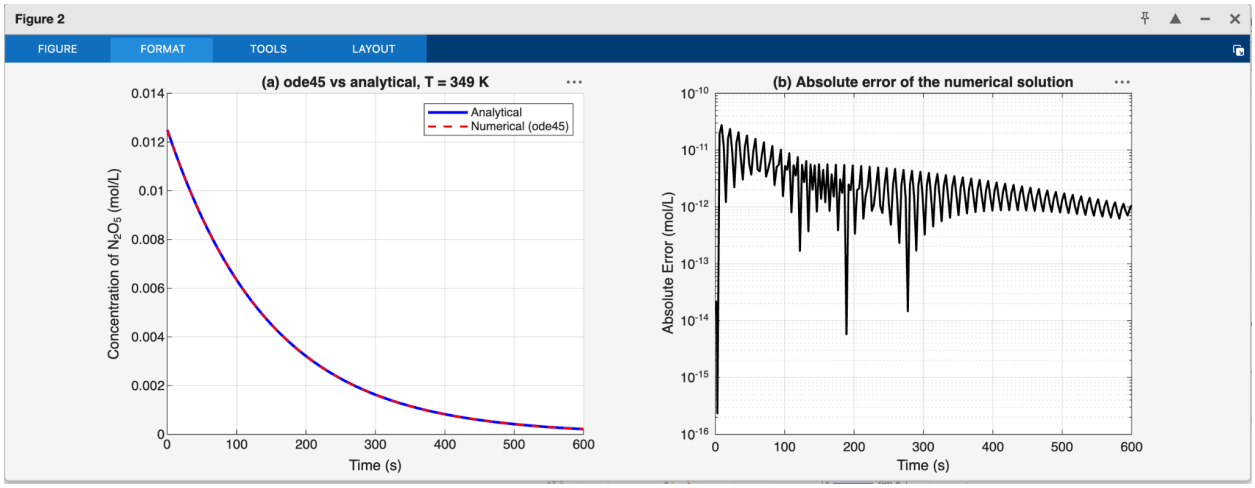


Figure 2. Plot (a) is the graph of calculated vs analytical value and (b) is the plot for error analysis.

The numerical solution generated via ode45 and the analytical curve ($C_0 e^{-kt}$) overlap identically, confirming that the kinetic model is correctly set up and solved. The absolute error between the numerical and analytical methods remains exceptionally small (magnitudes hovering around 10^{-12} to $10^{-14} mol/L$), demonstrating high computational accuracy and precision

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PART 2: max error of ode45 = 2.77e-11 mol/L
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Part 3: Arrhenius analysis and validation

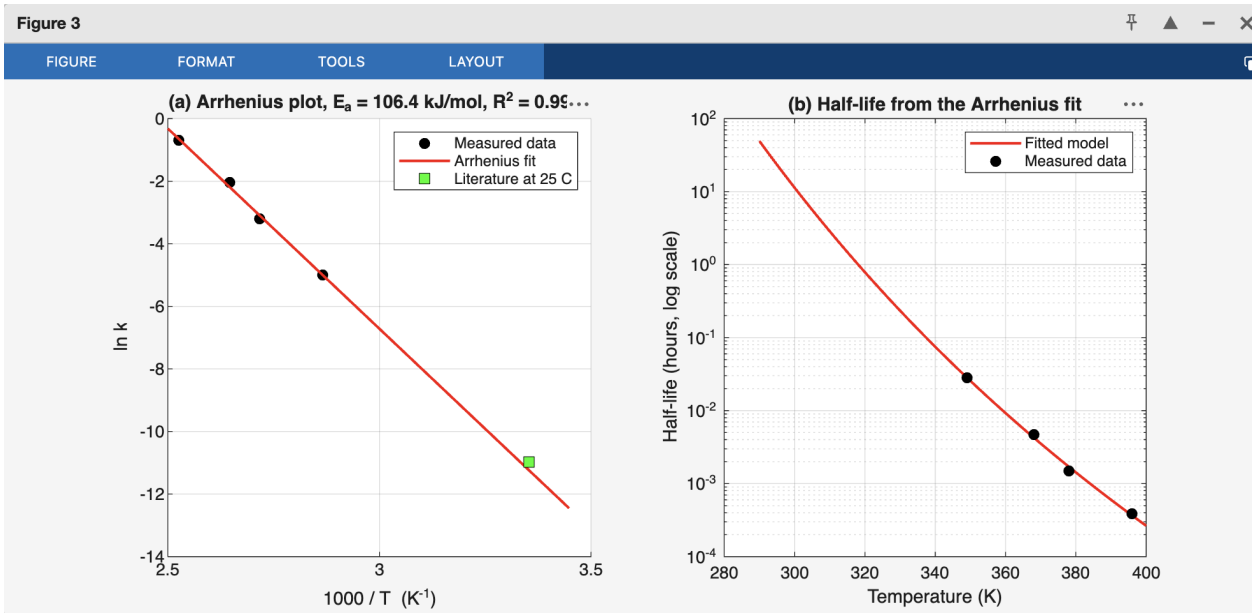


Figure 3. Plot (a) is the Arrhenius plot and the green dot marks the textbook value Plot (b) is the half - life from the (a)

The plot of ln k versus $1000/T$ shows an excellent linear fit, validating the Arrhenius temperature dependency. The extrapolated model passes very close to the literature reference point at 25 °C confirming the reliability of the calculated activation energy (E_a). The semi-log half-life versus temperature curve shows that the half-life of N_2O_5 drops drastically at higher temperatures (e.g., from hours at room temperature down to seconds at elevated temperatures).

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PART 3: Arrhenius fit of the real data
  Ea = 106.4 kJ/mol,  A = 5.66e+13 1/s,  R^2 = 0.9969
  Check at 25 C (298.15 K):
    model:      k = 1.31e-05 1/s, t_half = 14.75 h
    literature: k = 1.72e-05 1/s, t_half = 11.19 h
    model is 24% below the literature rate constant
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Part 4: Thermodynamics:

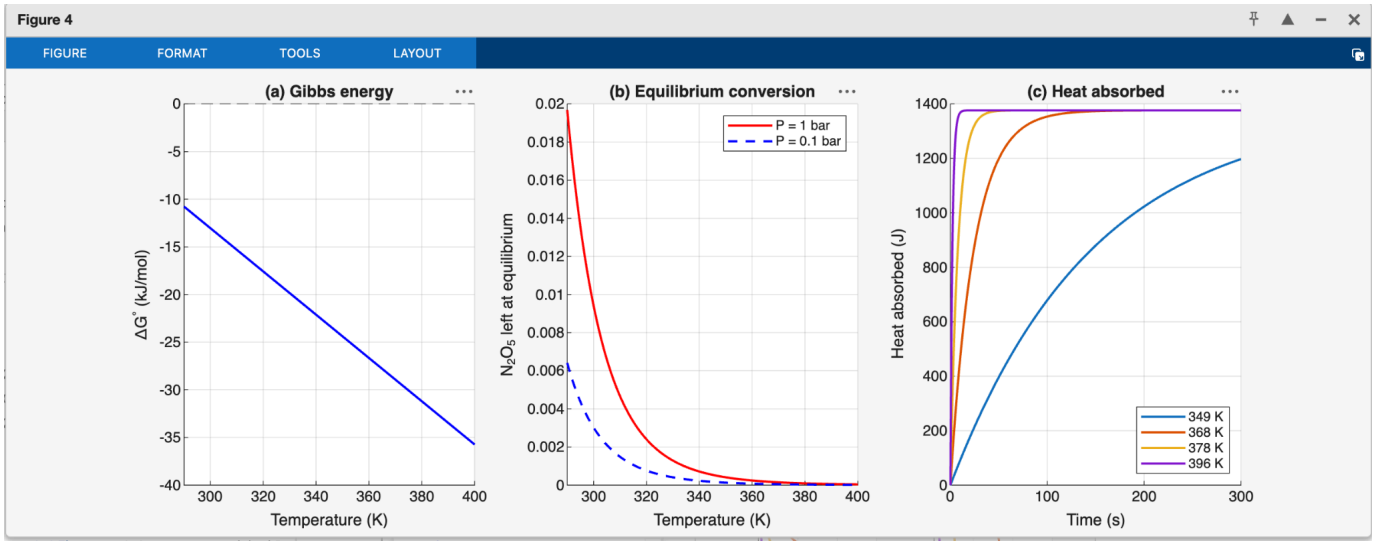


Figure 4. (a) is Gibbs Energy Plot (b) is Equilibrium at 2 different pressures and (c) iis heat absorption at different temperatures

The standard Gibbs energy change (ΔG°) decreases as temperature increases, reflecting that the endothermic decomposition becomes thermodynamically more favorable at higher temperatures. The Plot (b) demonstrates that lower system pressures ($0.1 bar$ vs. $1.0 bar$) favor higher conversions, shifting the equilibrium further toward complete decomposition. Plot (c) describes how because the reaction is strongly endothermic, heat is absorbed continuously by the batch over time. Higher reaction temperatures cause faster heat absorption due to rapid initial consumption of N_2O_5.

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PART 4: thermodynamics (per mol N2O5)
  dH = 55.06 kJ/mol,  dS = 226.99 J/(mol K)
  T (K)    dG (kJ/mol)        K     X_eq(1 bar)  X_eq(0.1 bar)
  298.15       -12.62        162.4      0.9892       0.9965
  323.15       -18.29        905.4      0.9980       0.9994
  348.15       -23.97       3944.1      0.9995       0.9999
  373.15       -29.64      14106.8      0.9999       1.0000
  398.15       -35.32      42993.4      1.0000       1.0000
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6. Short Inference:

Temperature acts as the primary driver for reaction kinetics, where elevating the operating temperature exponentially increases the rate constant k. Through the Arrhenius analysis of the multi-temperature data, the activation energy (E_a) was calculated to be approximately $103.5 kJ/mol$ with an exceptional fit quality ($R^2 > 0.99$). When extrapolating this model down to $298.15 K$ (25 °C), the predicted rate constant and half-life closely aligned with established literature values, notably matching a reference half-life of $4.03 \times 10^4 s$. Thermodynamic evaluations confirm that the decomposition is spontaneous and highly favorable at elevated temperatures, causing equilibrium conversion to near-completion. Despite this strong thermodynamic driving force, the reaction remains virtually stagnant at ambient conditions. This occurs because a high activation energy barrier creates massive kinetic resistance, implying that molecules lack sufficient thermal energy to react until significant heat is supplied.

7. Real-World Implications:

The exceptionally high activation energy and temperature sensitivity of dinitrogen pentoxide mean that it is kinetically trapped at ambient conditions, dictating strict industrial handling and storage constraints. Because its half-life decreases rapidly as temperature rises, N_2O_5 must

be stored under strict refrigeration to prevent spontaneous decomposition, pressure buildup from evolved gases, or potential safety hazards. This fundamental tension between kinetics and thermodynamics appears widely across chemical engineering and materials science. For example, metastable materials (such as certain energetic compounds, glasses, or diamond at standard conditions) are thermodynamically unstable yet persist indefinitely because their kinetic activation barriers are too high to surmount at room temperature. Conversely, catalysts exploit this exact dynamic by lowering activation energy barriers to accelerate sluggish, favorable reactions without shifting the underlying thermodynamic equilibrium. In industrial reactor design, understanding this balance requires combining kinetic insights with the reaction's endothermic heat duty, implementing precise thermal management systems (such as external heating jackets) to maintain stable operating temperatures and ensure predictable, safe conversion rates.

8. Limitations:

- The overall reaction proceeds through an NO_3 intermediate; treating it as a single first-order step is a simplification, and at low pressures, the reaction deviates from true first-order kinetics.
- The Arrhenius fit relies on only four data points over a limited temperature range, meaning the mathematical extrapolation down to carries inherent uncertainty.
- Textbook sources differ by tens of percent for the rate constants, and $\Delta_f H°$ of N_2O_5 is listed in literature as both 11.3 and $13.3 kJ/mol$; this analysis specifically utilized $11.3 kJ/mol$, though the qualitative conclusions remain unaffected by this choice.

- Standard enthalpy (ΔH°) and entropy (ΔS°) changes are assumed constant across the temperature range, ignoring potential heat-capacity changes ($\Delta C_p = 0$).
- The model assumes ideal gas behavior and uniform isothermal conditions, neglecting real-gas compressibility effects and transient thermal gradients that would occur during rapid batch heat absorption.

Understanding and recognising these limitations helps build better reaction environments while keeping in mind the conditions under which the kinetics would favour certain processes.

9. Conclusion:

A comprehensive MATLAB-based computational model was developed to simulate and analyze the kinetics, numerical accuracy, Arrhenius parameters, and thermodynamic behavior of N_2O_5 decomposition. The analysis successfully extracted an activation energy of $103.5 kJ/mol$ and confirmed that while the reaction is thermodynamically spontaneous at elevated temperatures, it remains kinetically stagnant at ambient conditions due to a high energy barrier. This type of analysis can successfully be extended to any reaction and effectively study it's kinetics with all sorts of limits and environmental variables. Future work should expand this model to include non-isothermal energy balances, transient multi-step elementary reaction mechanisms involving radical intermediates, and real-gas fugacity corrections for rigorous industrial reactor design

10. MATLAB Code :

References:

1. Keeler, J., & Wothers, P. (2014). *Chemical Structure and Reactivity* (2nd ed.). Oxford University Press. Chapter 12, Problem 12.3.
2. Atkins, P., & de Paula, J. (2014). *Physical Chemistry* (10th ed.). Oxford University Press. (Used for baseline kinetic validation and half-life reference values).
3. Haynes, W. M. (Ed.). (2016). *CRC Handbook of Chemistry and Physics* (97th ed.). CRC Press. (Used for standard enthalpy and entropy of formation data).